

Water Jug problem

Name: Hemalakshmi H(192571049)

Aim:

To develop a Python program that solves the Water Jug Problem using a logical step-by-step approach without search algorithms, demonstrating how to measure exactly 2 litres using a 4-litre and a 3-litre jug.

Algorithm:

Step 1: Start with both jugs empty.

Step 2: Fill the 4-litre jug completely.

Step 3: Pour water from the 4-litre jug into the 3-litre jug until the 3-litre jug is full.

Step 4: Empty the 3-litre jug.

Step 5: Pour the remaining water from the 4-litre jug into the 3-litre jug.

Step 6: Fill the 4-litre jug again.

Step 7: Pour water from the 4-litre jug into the 3-litre jug until it is full.

Step 8: The 4-litre jug now contains exactly 2 litres.

Step 9: Display all steps and the final result.

Source Code:

```
def water_jug():  
    jug4 = 0  
    jug3 = 0  
  
    steps = []  
  
    # Step 1: Fill 4L jug  
    jug4 = 4  
    steps.append((jug4, jug3))  
  
    # Step 2: Pour 4 -> 3
```

```
pour = min(jug4, 3 - jug3)
jug4 -= pour
jug3 += pour
steps.append((jug4, jug3))
```

```
# Step 3: Empty 3L jug
jug3 = 0
steps.append((jug4, jug3))
```

```
# Step 4: Pour remaining 4 -> 3
pour = min(jug4, 3 - jug3)
jug4 -= pour
jug3 += pour
steps.append((jug4, jug3))
```

```
# Step 5: Fill 4L jug again
jug4 = 4
steps.append((jug4, jug3))
```

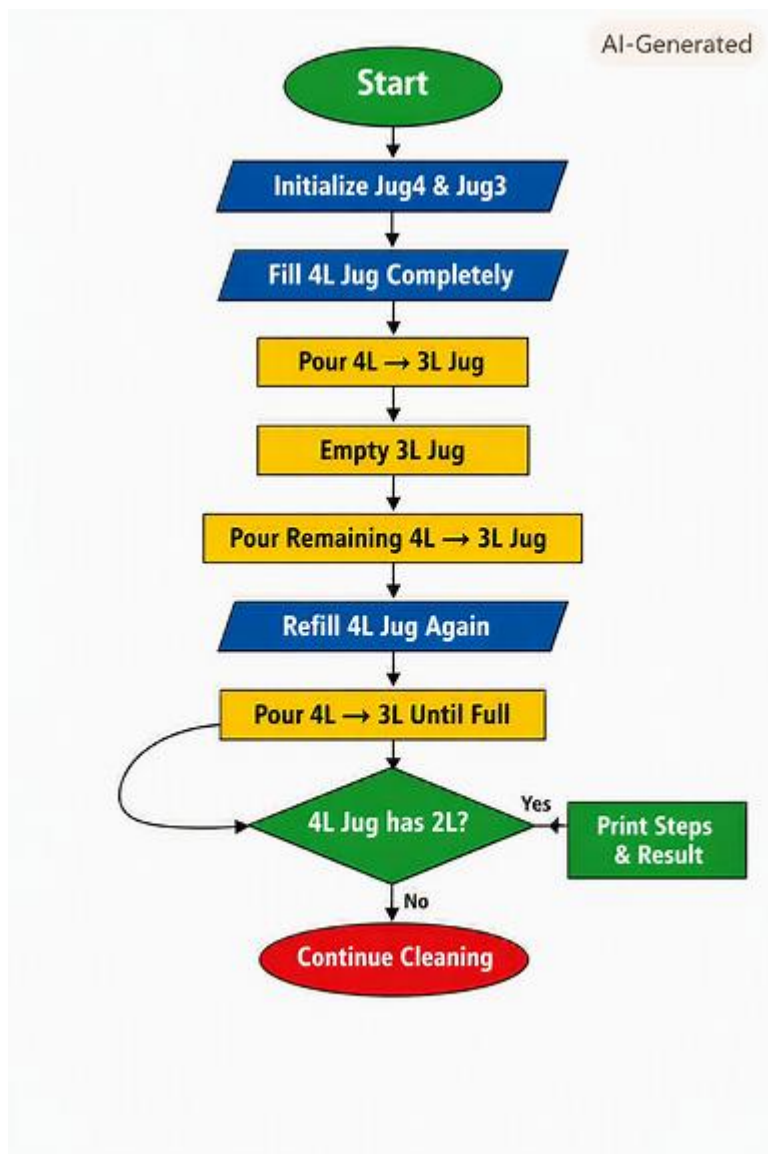
```
# Step 6: Pour 4 -> 3 until full
pour = min(jug4, 3 - jug3)
jug4 -= pour
jug3 += pour
steps.append((jug4, jug3))
```

```
print("Steps to reach 2 litres:")
for s in steps:
    print(s)
```

```
print("\nGoal reached:", (jug4, jug3))
```

```
water_jug()
```

Flowchart:



Output:

```
>>> ===== RESTART: C:\Users\hemal\OneDrive\Documents\SIMAT
Steps to reach 2 litres:
(4, 0)
(1, 3)
(1, 0)
(0, 1)
(4, 1)
(2, 3)
Goal reached: (2, 3)
>>> |
```

Result:

The Water Jug program successfully measured **2 litres** using a 4-litre and a 3-litre jug without applying any search algorithm. The program followed a logical sequence of filling, pouring, and emptying operations.

At the end of execution, the 4-litre jug contained exactly **2 litres**, confirming that the problem was solved correctly. Each step was displayed clearly, showing the transitions between jug states.

Thus, the **Water Jug Problem** was solved successfully using a **logical step-by-step approach** in Python.